

COURSE DESCRIPTION			
Program	Commercial Practice		
Course Title	Data Entry Operations and Applications		
Course Code	2148	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

A data entry operation involves inputting, updating, and managing information into computer systems, databases, or spreadsheets, primarily converting physical, audio, or raw data into digital formats.

Course Objectives

1	To provide hands-on experience in Data Entry Operations.
2	To impart skills in various spreadsheet operations.
3	To familiarize with the power point creation and presentation
4	To familiarize the students with the fundamentals of computer operating system.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Fundamental knowledge in formatting and editing documents.	2149	Typewriting and Word Processing I	I
Typing speed using key board.	2149	Typewriting and Word Processing I	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply the fundamental knowledge of computer	Apply
CO2	Classify user interface of MS Excel, work book	Apply
CO3	Formatting Worksheets, Formulas, Functions, pivot table & charts	Apply
CO4	Creating Power Point Presentation	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	3	-	-	-	-	-	-
CO2	3	-	-	-	3	-	-	-	-	-	-
CO3	3	3	2	-	3	-	-	-	-	-	-
CO4	2	-	2	-	2	-	-	-	3	-	-
Total	11	3	4	-	11	-	-	-	3	-	-
Correlation Level	2.75	3	2	-	2.75	-	-	-	3	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Computer fundamentals	Illustrate the features and application of system hardware and software	CO1	Apply	2
2	Input and output devices	Classify different Input Output devices	CO1	Apply	1
3	Memory	Examine primary memory & secondary memory	CO1	Apply	1
4	Software	Expose system software & application software	CO1	Apply	1
5	Operating system	Examine functions of operating system	CO1	Apply	1
6	Windows/ Linux	Discover advantages and application of Windows, Installation procedure for operating systems various application	CO1	Apply	3
Module II					
7	User interface	Explore MS Excel, Work book	CO2	Apply	2
8	Google drive and cloud folders	Operate Google drive and cloud folders	CO2	Apply	1
9	Google drive and cloud folders	Exercise File sharing permission	CO2	Apply	1
10	Intelligent Document Processing	Explore IDP, Automatic data extraction, extract key fields	CO2	Apply	2
11	AI based document management	Discover document workflow and productivity - convert scanned document into OCR	CO2	Apply	2
12	Tool bar	Project dynamic tool bar	CO2	Apply	1
13	Formatting tool bar	Process Cell formatting. Practice resizing of rows and columns	CO2	Apply	2
14	Working with formula	Use arithmetic, comparison, reference	CO2	Apply	1
15	Working with formula	Illustrate Fill series, Sum and average, If - If else	CO2	Apply	2
Module III					
16	Utilize function	Apply the function date, count, min, max	CO3	Apply	1
17	Utilize function	Operate with Concatenate, upper, lower, today, Len, Now	CO3	Apply	2
18	Range	Employ working with range	CO3	Apply	1
19	Print area	Assign Print area and print	CO3	Apply	1
20	Pivot table	Construct a pivot table	CO3	Apply	2
21	Working with Chart	Draw Pie chart, column chart, line charts and other charts	CO3	Apply	2
Module IV					
22	MS Power point	Explore the features	CO4	Apply	1
23	Powerpoint presentation	Prepare presentation using both blank template and installed	CO4	Apply	2
24	Slide transition with effects and sound	Show Slide transition with effects	CO4	Apply	1
25	Slide transition with effects and sound	Discover designing of slides	CO4	Apply	1
26	Slide transition with effects and sound	Construct Layout slides	CO4	Apply	2
27	Slide transition with effects and sound	Create sound effect	CO4	Apply	1
28	Animation	Show Custom animation and slide show	CO4	Apply	2
29	Create presentation	Prepare Ads/Projects/Seminars	CO4	Apply	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.

Integrate multiple concepts from the syllabus into a combined application.

Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.

Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare Charts with pictures showing input and output devices	1.5
2	Compare Primary and Secondary memory	1.5
3	Prepare the history of operating systems	1.5
Module II		
4	Create a folder and upload it to your google drive	1.5
5	Apply OCR using any IDP tool or OCR software to convert PDF	1.5
6	Extract fields from any PDF file and save the extracted data in Excel.	1.5
7	Calculate sum and average mark of your semester 2 result using excel	1.5
8	Prepare for the lab exam	1.5
Module III		
7	Create a Time Table using excel, take a print.	1.5
8	Create a data sheet of your class mates including name, age, DoB, name of parent, mobile number etc.	1.5
9	Create attendance sheet of your class for last month including sum, average, if, if - else, max, min etc	1.5
10	Independently attempt to create a pivot table using assuming data	1.5
11	Create various charts using first semester result of your whole class	1.5
Module IV		
12	Apply different slide transitions to your presentation	1.5
13	Redesign one slide in three different styles	1.5
14	Animate text, images, and shapes to your presentation	1.5
15	Prepare for the lab exam	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Computer Fundamentals with MS Office Applications	N Saravanan & D. Shanthi.	SCITECH Publications.
2	Exploring Computer Fundamentals MS office.	Navaneeth Mehra & Anupam Jain.	SCITECH Publications.

REFERENCE			
Sl. No.	Title	Author	Publication
1	Microsoft office fundamentals.	Laura Story & Dawna Walls	South-Western
2	Mastering MS Office..	Bittukumar	V& S publishers

WEB RESOURCES			
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Sl. No.	URL	Modules Covered
1	https://www.homeandlearn.co.uk/	All
2	https://edu.gcglobal.org/en/subjects/office/	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Desktop system	Intel Core i5/AMD Ryzen 5, 16GB RAM (8GB minimum), and a 512GB SSD, running Windows with a 24-inch Full HD monitor	30
2	Internet Access	64 Mbps (Minimum)	30
3	Printer	Epson Ecotank L3260 A4 Wi-Fi All-in-One Ink Tank Colour Printer, Black/any other	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	M S Excel	Latest	30
2	M S Power point	Latest	30

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to			15	15		
Marks	5	30	15		10	40
Total marks		60				40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8

Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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